

ECE444 PRA2 Front End Design

Goal

In this PRA, you will modify and deploy a personal website using HTML, CSS, and JavaScript. You will practise:

- modifying the content and appearance of an existing website;
- embedding external content;
- manipulating the DOM with JavaScript;
- deploying a website with GitHub Pages; and
- using Git commits to document your development process.

You may use an AI coding assistant. Regardless of how you complete the work, you are responsible for testing your website and understanding all submitted changes.

Starting Repository

Use the following repository as your starting point:

<https://github.com/varadbhogayata/varadbhogayata.github.io>

Commit Requirements

Complete the activities in order. Commit and push your work after each activity before beginning the next one.

Begin each commit message with the activity number. For example:

Activity 3: Change website colour theme

You may make multiple commits for an activity. Your commit history must make it clear which changes belong to each activity.

Activity 1 Create Your Repository

1. Create an empty public GitHub repository named <your-GitHub-username>.github.io.
2. If you already have a repository with that name, create one named ece444-pra2.
3. Clone the starting repository and push its complete codebase to your new repository.
4. Add your name to README.md.
5. Add the following acknowledgement to README.md:
This repository is based on
<https://github.com/varadbhogayata/varadbhogayata.github.io>.
6. Commit and push your changes.

Activity 2 Personalize and Deploy the Website

1. Replace the existing content in the About, Experience, Projects, Skills, Education, Contact, and Resume sections.
2. Use only information that you are comfortable publishing in a public repository.
3. Deploy the website using GitHub Pages.
4. Confirm that the deployed website is accessible at one of the following URLs:
 - a. <https://<your-GitHub-username>.github.io>
 - b. <https://<your-GitHub-username>.github.io/ece444-pra2/>
5. Commit and push your changes.

Activity 3 Change the Website Theme

1. Replace the existing green theme with a different color theme.
2. Apply the new theme consistently across the website.
3. Ensure that text remains easy to read against the selected colors.
4. Confirm that the website still displays correctly on GitHub Pages.
5. Commit and push your changes.

Activity 4 Add a Places Travelled Section

1. Create a section titled **Places Travelled**.
2. Add a link to this section in the website's navigation menu.

3. Embed a Google map showing one or more places you have visited or would like to visit. You are not required to disclose your actual travel history.
4. Confirm that the map loads correctly on the deployed website.
5. Commit and push your changes.

Activity 5 Add Dynamic Project Content

1. Store information for at least four projects in a JavaScript array.
2. Add a `<div>` with a unique `id` for displaying the projects.
3. Write a JavaScript function that reads the project data and inserts the corresponding HTML into the page.
4. Initially display two projects.
5. Add a **Load More** button that displays the remaining projects without reloading the page.
6. Hide or disable the button after all projects have been displayed.
7. Confirm that the feature works on the deployed website.
8. Commit and push your changes.

Activity 6 Add an Enhancement of Your Choice

Add one meaningful enhancement that is not already required by Activities 1–5. The enhancement must be visible or testable on the deployed website.

Possible enhancements include:

- improving the website's mobile layout;
- adding a dark mode;
- adding form validation;
- adding project filtering or searching;
- adding an animation or interactive component;
- improving keyboard navigation or accessibility; or
- adding a Bootstrap component.

You may choose a different enhancement. Updating text or changing colours alone does not satisfy this activity.

Test the enhancement on the deployed website, then commit and push your changes. Use a commit message beginning with **Activity 6**.

During grading, be prepared to demonstrate the enhancement and explain:

- why you selected it;
- how you implemented it; and
- how you tested it.

The enhancement is graded for correct implementation and demonstrated understanding, not for its complexity.

Submission

Submit the following on Quercus:

- your GitHub repository URL; and
- your deployed website URL.

Screenshots are not required.

TA Evaluation

After completing all activities, open your deployed website and GitHub commit history for the TA.

The TA will:

- Verify that the required features work;
- Select one or more commits from your history;
- Ask you to explain the changes made in those commits; and
- Ask questions about the relevant HTML, CSS, or JavaScript.

This PRA is worth 5 points:

- Completion and correctness of Activities 1–6: 3 points
- Explanation of selected commits and demonstrated understanding: 2 points

Resources

- [GitHub Pages](#)
- [CSS selectors](#)
- [CSS syntax](#)
- [CSS property reference](#)
- [JavaScript guide](#)
- [DOM manipulation](#)
- [Embed a Google map](#)